



TECHNICAL UNIVERSITY

OF CLUJ-NAPOCA, ROMANIA

FACULTY OF AUTOMATION AND COMPUTER SCIENCE

Real-Time Light Dosimetry for Heritage Digital Twins via Texture Space Ray Tracing

LICENSE THESIS

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Real-Time Light Dosimetry for Heritage Digital Twins via Texture Space Ray Tracing

1. **Project proposal:** Development of ChronoLux, an interactive Digital Twin for real-time light dosimetry using Texture Space Ray Tracing in DirectX 12.
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Chapter 1. Introduction

1.1. Context and Motivation

The concept of cultural heritage conservation is one that is highly complex, being highly scientific and involving the balancing act between access to the public and the protection of artifacts. In museums, it is common practice to face a situation where there is a need to display old paintings, which could be as old as 16th century oil paintings, delicate pieces of cloth and even water colors without damaging them. The most ubiquitous source of degradation of such artifacts, however, is photochemical degradation caused by the presence of light.

Light Dosimetry is the metrological way of quantifying the radiative energy, measured in Lux Hours, acting upon an artifact within a specific period. Given that this damage is strictly cumulative and irreversible, it becomes imperative that preventive conservation measures be taken. For this purpose, guidelines set forth by international bodies like the International Commission on Illumination (CIE) are used to govern preventive conservation of artifacts. For highly susceptible items, the recommendation made by CIE is that the maximum limit of energy to which the artifact may be exposed should not exceed 50,000 Lux Hours annually.

1.2. Limitations of Existing Solutions

Up to now, the management of the stress induced by the light had been relying on two separate paradigms. However, both have their respective limitations when applied to today's workflows in the conservation of valuable heritage.

According to the first approach, physical light measurement can be carried out by the means of IoT sensor networks and photometers. While these devices indeed provide the essential ground truth data, they cannot help predict a threat before its occurrence; therefore, there is always going to be some delay, which makes these sensors unable to provide any proactive measure prior to a new exhibition. Besides, while measuring irradiance at a single point in space, it would take a huge amount of sensors deployed at high density in order to fully cover all possible angles of light incidence in a large object. Not only it would be expensive and unattractive, but mounting these devices on fragile pieces would simply cause irreversible damage. Finally, as every other device with a physical sensor inside, it suffers from calibration error, which needs to be compensated for by regular maintenance.

According to the second one, the issue of predicting is approached by using offline rendering engines like Radiance or Precomputed Radiance Transfer (PRT) lightmaps, which calculate the lighting distribution mathematically and in an extremely reliable way. Yet their inability to exploit hardware acceleration capabilities and being completely CPU-bound renders them impractical for real-time applications due to their poor

performance, which results in hours' worth of calculation just for a single, one-year exposure simulation done with hourly precision (which is required in order to achieve proper global illumination).

That is precisely why this offline approach does not allow any interactive exploratory process to be carried out, meaning that conservators can neither explore the possibility of introducing some interventions into the environment—like altering transmittance of window glass, changing room albedo or moving objects around—nor try some optimal combination thereof.

1.3. The Proposed Solution: ChronoLux

For mitigating the issue of the interactivity and fidelity gap, this research will introduce ChronoLux - an interactive Digital Twin platform designed specifically for near-real-time dosimetry. Making use of modern GPU acceleration capabilities through DirectX 12 Raytracing (DXR), ChronoLux will transform passive artifact three-dimensional representations into interactive, predictive, and accurate simulations.

As opposed to camera-oriented screen-space rendering through camera-to-image ray-casting performed by current generation interactive graphics platforms, ChronoLux introduces a novel architectural approach - the Texture Space Ray Tracing. Regular renderers conduct frustum culling to exclude rays that are cast beyond the view range of the camera. Such an approach is insufficient when simulating light interaction for scientific dosimetry since measurements need to be conducted persistently throughout the entire geometrical surface of an object. Instead, using the artifact's UV coordinates, ChronoLux will cast rays from those points thus baking any cumulative exposure permanently into the object's surface.

In order to guarantee accuracy and metrological precision, the simulation will run under the Perez All-Weather Sky Model that will provide exact positioning of sunlight along with the skylight irradiance. The core of the algorithm will use the stochastic Next Event Estimation (NEE) Monte Carlo path tracer that will allow sampling separate components of illumination such as direct solar beams and diffuse scatterings independently. Overall, this approach provides curators with a non-destructive sandbox environment that enables instant dosimetric measurements without any drop in performance at approximately 60 fps rate.

1.4. Thesis Structure

Subsequently, the body of this paper is arranged in such a way that all the necessary information regarding ChronoLux will be delivered, from theory to experimental confirmation.

In Chapter 2, the main goals of this study will be described, namely, both interactive goals and metrological ones. Chapter 3 represents an extended review of the bibliography and describes recent achievements in the areas of physical dosimetry, offline ray tracing, and hardware accelerated global illumination. Chapter 4 contains the analysis of the theory of light transport, including the explanation of Lambert's Cosine Law, and Monte Carlo integrals. Chapter 5 describes the process of creation of the Digital Twin and focuses on the Texture Space Ray Tracing algorithm, as well as Next Event Estimation kernel. In Chapter 6, the experimental validation and testing of the proposed

framework is described in detail, including different conservation case studies and proving the mathematical consistency of the Monte Carlo convergence. In addition, Chapter 7 presents an elaborate user guide to using the software, and in Chapter 8 - conclusions and future perspectives are stated.

Chapter 2. Project Objectives

The main focus of this thesis is to create a hardware-accelerated Digital Twin, called *ChronoLux*, which will provide an interactive and physically-based lighting simulation for the conservation of cultural heritage. This solution is designed to fill the important missing link between computationally-heavy offline rendering systems (such as Radiance), which can be proven mathematically, and real-time IoT sensing devices.

In order to achieve this high-level goal and provide the necessary combination of interactivity and scientific accuracy, the development process is carried out based on certain technical and metrological goals. The high-level architecture related to these goals is outlined in Figure 2.1.

2.1. Technical and Architectural Objectives

- 01. Texture Space Ray Tracing Integration:** To address the drawbacks associated with traditional screen space algorithms, such as frustum culling that results in critical out-of-screen lighting information being removed from calculations, a special compute shader for DirectX 12 Raytracing (DXR) would have to be used to launch rays directly from the unwrapped UV texture map of 3D models in order to permanently bake the cumulative radiometric energy into all of its geometric surface regardless of the location of the observer.
- 02. Physical Environmental Modeling:** In order to combine the Perez All-Weather Sky Model with a special Next Event Estimation (NEE) compute algorithm to create a scientifically rigorous method to decouple the direct beams of sunlight from the diffuse ambient lighting, thus generating physically correct global illumination without loss of real-time processing ability.
- 03. Asynchronous Compute Architecture:** For a proper separation of concerns between the user interface layer and computationally demanding GPU path tracing engine, the program should allow users to modify physical properties (transmission coefficients of window glazing or wall reflectance, for instance) via an interactive UXML/USS user interface without breaking the continuity of the 60 FPS viewport interaction.

2.2. Metrological and Validation Objectives

- 04. Headless Statistical Telemetry:** In this study, an automated approach has been developed for virtual point-probe Lux sensors which operate purely in the background. The sensors are created for the purpose of computing spatial variance (σ^2), maximum exposure dose, and standard Monte Carlo convergence error boundaries while not interfering with the primary rendering process.

O5. Empirical Scenario Validation: Explicit validation of the above approach is achieved by means of running simulations of an entire year of meteorology (i.e., 8,760 hours of continual processing) for various preventive conservation scenarios. It is essential for establishing the efficacy of dynamic environmental control actions like the application of UV-absorbent windows or optimizing geometric occlusion of susceptible artworks.



Figure 2.1: High-Level Concept Architecture of the ChronoLux framework, illustrating the pipeline from raw heritage asset ingestion to real-time dosimetric data export.

The subsequent chapters of this thesis will methodically detail the design, theoretical foundation, programmatic implementation, and scientific validation of these stated objectives.

Chapter 3. Bibliographic Research

The development of a hardware-accelerated Digital Twin for real-time light dosimetry exists at the intersection of several complex scientific domains. This chapter provides a rigorous review of the state-of-the-art literature encompassing preventive conservation metrology, mathematical global illumination, and modern graphics hardware architectures. By analyzing existing methodologies, we establish both the scientific foundation of this thesis and the critical limitations of current tools that *ChronoLux* seeks to resolve.

3.1. Preventive Conservation and Empirical Dosimetry

The fundamental necessity of light dosimetry is rooted in the photochemical vulnerability of cultural heritage artifacts. As demonstrated by Michalski [1], both visible light and ultraviolet (UV) radiation induce irreversible molecular degradation in organic and synthetic materials. At a molecular level, photons carry energy inversely proportional to their wavelength. UV radiation (ranging from 10nm to 400nm) carries sufficient quantum energy to break the covalent bonds of pigments, binders, and structural fibers. This photochemical degradation manifests physically as color fading (bleaching), yellowing of varnishes, and structural embrittlement of textiles and paper.

Because photon-induced decay is strictly cumulative, damage cannot be reversed or "healed" by placing an artifact in the dark; the molecular bonds are permanently severed. To systematically manage this risk, the International Commission on Illumination (CIE) established strict preventive conservation limits and categorized materials into distinct vulnerability classes [2]:

- **Insensitive Materials:** Metals, stone, and ceramics. These materials do not suffer photochemical degradation and require no strict lighting limits.
- **Low Sensitivity Materials:** Oil and tempera paintings, frescoes, and structural woods. These can tolerate moderate exposure, typically limited to 600,000 Lux-Hours annually.
- **High Sensitivity Materials:** Watercolors, pastels, tapestries, and antique paper. These artifacts are extremely vulnerable and are strictly capped at 50,000 Lux-Hours annually.
- **Extreme Sensitivity Materials:** Silk, specific unstable organic dyes, and early photographs. These items demand maximum protection, often restricted to 15,000 Lux-Hours annually.

Exceeding these precise thresholds accelerates the entropy of the artifact, necessitating highly accurate prediction mechanisms to design safe exhibition spaces.

Historically, the dominant methodology for tracking this exposure has been the physical measurement of incident light. Silva et al. [3] highlight the usage of low-cost data loggers to monitor the microclimate of heritage sites. Furthermore, Chianese et al. [4] propose the concept of the "Internet of Cultural Things," advocating for dense IoT

sensor networks embedded within the museum infrastructure.

Discussion: While empirical sensor networks provide undeniable ground-truth data, they suffer from critical operational flaws that severely restrict their utility in modern conservation. First, they are fundamentally reactive. A physical sensor only triggers an alert after the light exposure has already occurred, offering absolutely no predictive capabilities to prevent damage before a new exhibition begins. Second, capturing the complex spatial variance of irradiance across a large, three-dimensional geometric surface requires deploying an impossibly dense grid of physical sensors. Deploying thousands of physical probes across a Renaissance statue is not only financially prohibitive but aesthetically intrusive, and mounting them on delicate surfaces poses an unacceptable structural risk. Ultimately, to transition from reactive measurement to proactive, non-destructive prediction, researchers must utilize virtual, mathematically driven simulations of light transport.

3.2. The Metrology of Global Illumination

3.2.1. Rasterization vs. Ray Tracing in Scientific Simulation

In the domain of computer graphics, visualizing virtual environments has historically been dominated by rasterization algorithms. Rasterization operates by projecting 3D polygons onto a 2D screen space and determining visibility through depth buffering. While incredibly fast and well-suited for interactive video games, rasterization is fundamentally incapable of scientific dosimetry. Because rasterization only computes direct visibility from the camera’s perspective (frustum culling), it relies heavily on “hacks” and approximations to simulate indirect light—such as Screen Space Ambient Occlusion (SSAO), shadow mapping, and baked light probes. These techniques are heavily biased and provide only an illusion of global illumination.

Conversely, ray tracing explicitly models the physical trajectory of photons as they travel from a light source, intersect with geometry, and reflect or refract based on the physical material properties. Historically, the primary use case for ray tracing has been the generation of stunning, photorealistic graphics. Because it inherently calculates complex optical phenomena such as perfect mirror reflections, soft shadows, and transparent refractions, it has been the gold standard for visual effects (VFX) in cinema and, more recently, high-end video games. However, in these commercial applications, the ultimate goal is merely to deceive the human eye into perceiving realism, often cutting corners on physical accuracy to save rendering time.

Scientific dosimetry requires strict adherence to physical light transport, independent of a virtual camera or human perception. To achieve this, the simulation must rely on mathematically pure, unbiased ray tracing. Rather than generating a visually pleasing pixel on a screen, the path-traced photons must act as perfect radiometric data carriers. This approach is mathematically provable, making it the only viable algorithmic paradigm for measuring real-world irradiance and calculating precise Lux-Hour accumulations in a virtual space.

3.2.2. The Rendering Equation and Monte Carlo Integration

The foundational methodology for physically based ray tracing is the Rendering Equation, formalized by Kajiya [5]. This integral equation mathematically describes the

equilibrium of light transport within a scene, stating that the total outgoing radiance from a surface point is the sum of emitted radiance and the integral of incoming radiance scattered over the hemisphere.

To solve this infinite-dimensional integral, graphics engines employ Monte Carlo integration. Because it is computationally impossible to evaluate an infinite number of incoming light directions, Monte Carlo methods estimate the integral by casting a finite, random set of rays (samples). The accuracy of this estimation increases as the number of samples approaches infinity, a concept governed by the Law of Large Numbers.

However, uniformly shooting random rays is incredibly inefficient, as most rays will fail to hit a significant light source, resulting in a noisy image (high variance). To solve this, importance sampling is utilized. By defining a Probability Density Function (PDF) that matches the expected distribution of light (e.g., Cosine-Weighted Importance Sampling for Lambertian surfaces), the algorithm preferentially casts rays in directions most likely to contribute significant energy.

To ensure that rays do not bounce infinitely inside enclosed spaces, stochastic termination techniques such as Russian Roulette are employed. Instead of enforcing an arbitrary hard cap on the number of bounces, Russian Roulette terminates a ray probabilistically based on the physical albedo of the surface it strikes. Darker surfaces, which absorb more light, have a higher probability of terminating the ray path, ensuring that computational resources are not wasted tracking photons whose energy has already decayed to negligible levels.

3.2.3. Physical Material Simulation and Energy Conservation

The physical accuracy of a global illumination simulation relies entirely on how mathematically strict its material properties are defined. When a photon intersects a geometric surface, its subsequent behavior is dictated by the material's Bidirectional Reflectance Distribution Function (BRDF) and Bidirectional Transmittance Distribution Function (BTDF). Together, these functions describe how incoming irradiance is scattered, absorbed, or transmitted across the hemisphere of the surface normal.

Modern computer graphics often rely heavily on complex Physically Based Rendering (PBR) materials. As detailed by Pharr et al. [6], PBR models utilize advanced microfacet equations (such as GGX distribution) to simulate the microscopic roughness and metallic properties of a surface to achieve extreme photorealism. However, evaluating these complex microgeometry equations introduces massive computational overhead that contributes very little to the calculation of pure radiometric energy dosage on flat architectural surfaces.

Consequently, *ChronoLux* intentionally eschews complex visual PBR in favor of a computationally streamlined, strictly thermodynamic solution. In scientific metrology, materials must adhere to the First Law of Thermodynamics (Energy Conservation). This principle dictates that the sum of a material's Reflectance (R), Transmittance (T), and Absorptance (A) must exactly equal 1.0 ($R + T + A = 1.0$). If a theoretical material reflects 60% of incoming light and transmits 30% (such as a tinted window glass), it mathematically must absorb the remaining 10%.

Simulating these properties is critical for preventive conservation. For example, structural walls generally exhibit Lambertian diffuse reflectance, scattering photons equally in all directions, which drastically influences the ambient indirect lighting of a room. Conversely, window glazing exhibits complex specular transmittance and refrac-

tion. Accurately modeling the transmittance curve of UV-blocking glass or the high reflectance of a polished marble floor is mandatory, as these material properties fundamentally dictate the distribution and intensity of the accumulated Lux-Hour dosage on the vulnerable artifacts. By enforcing a strict $R + T + A = 1$ model using pure Lambertian and ideal specular distributions, the engine guarantees metrologically valid dosimetry without the performance penalty of microfacet evaluation.

3.2.4. Limitations of Offline Rendering

The most prominent implementation of these principles in architectural and conservation analysis is the Radiance suite [7]. Radiance employs a backward, deterministic ray-tracing algorithm coupled with irradiance caching to solve the Rendering Equation with exceptional metrological accuracy.

Discussion: Radiance provides mathematically provable lighting distributions, making it a gold standard for offline dosimetry. However, its complete reliance on CPU-bound computations results in severe performance bottlenecks. Calculating a single year of cumulative exposure can take several hours, inherently preventing any real-time, interactive exploration of conservation strategies.

Alternative methodologies, such as Precomputed Radiance Transfer (PRT) [8], attempt to achieve real-time global illumination by pre-calculating light transport into spherical harmonics. While PRT offers high performance for static scenes, it completely fails the moment environmental interventions become dynamic. If a conservator wishes to rotate an artifact or change the transmittance of window glazing, the entire PRT lightmap must be recomputed from scratch.

3.3. Hardware Acceleration and Heritage Digital Twins

The synthesis of highly accurate dosimetry with real-time interactivity requires modern architectural paradigms. Jouan and Hallot [9], as well as Niccolucci et al. [10], define the "Heritage Digital Twin" not merely as a static 3D model (like traditional HBIM), but as a dynamic replica continuously updated with physical data to support preventive conservation policies.

3.3.1. Bounding Volume Hierarchies and Ray Traversal

Bringing the Rendering Equation into this interactive realm at 60 Frames-Per-Second requires advanced spatial acceleration structures. Computing the intersection of a ray against millions of raw geometric triangles is mathematically intractable for real-time execution. To solve this, scenes are partitioned using a Bounding Volume Hierarchy (BVH).

A BVH is a tree structure where geometry is grouped into Axis-Aligned Bounding Boxes (AABB). The root node contains the entire scene, and each subsequent child node progressively subdivides the geometry until the leaf nodes, which contain the actual triangles. When a ray is cast, it tests against the bounding boxes. If a ray misses a parent AABB, the algorithm instantly discards all millions of triangles contained within it, turning an $\mathcal{O}(n)$ intersection problem into a logarithmic $\mathcal{O}(\log n)$ search.

To optimally construct these trees, the Surface Area Heuristic (SAH) is employed. SAH minimizes the statistical probability of a ray hitting a bounding box by calculating

the cost of traversing nodes versus the cost of intersecting triangles, guaranteeing the most efficient BVH structure possible.

3.3.2. DirectX Raytracing (DXR) Architecture

Historically, constructing and traversing the BVH was entirely handled by software on the CPU. However, as detailed by Burgess [11], the introduction of the NVIDIA Turing architecture fundamentally shifted this paradigm by introducing dedicated hardware silicon—RT Cores—explicitly designed to accelerate BVH traversal and ray-triangle intersections.

This hardware acceleration is exposed to developers via the DirectX 12 Raytracing (DXR) API. DXR introduces a two-tier acceleration architecture designed to balance static and dynamic geometry. The Bottom-Level Acceleration Structure (BLAS) houses the raw triangle data for individual meshes and is highly optimized but computationally expensive to build. The Top-Level Acceleration Structure (TLAS) contains lightweight instance references to the BLAS, allowing developers to rigidly transform, rotate, or animate objects instantly without triggering a costly rebuild of the underlying triangle hierarchy.

DXR also introduces a modernized shader pipeline heavily utilized in frameworks like ChronoLux:

- **RayGeneration Shader:** The absolute entry point of the pipeline, responsible for invoking the primary rays. Instead of launching rays from a virtual camera, specialized frameworks launch them from Texture Space.
- **ClosestHit Shader:** Triggered when a ray successfully hits opaque geometry. This is where material characteristics, such as BRDFs, physical albedo, and diffuse interreflections, are evaluated.
- **AnyHit Shader:** Intercepts a ray before ClosestHit to evaluate transparency or alpha testing. This is crucial for scientific dosimetry when calculating physical transmittance through refractive window glazing without terminating the shadow ray.
- **Miss Shader:** Triggered when a ray escapes the local geometry entirely. In environmental simulations, the Miss shader queries global illumination models, such as the Perez All-Weather Sky Model [12], to calculate exact astronomical solar irradiance and ambient skylight.

3.3.3. Variance Reduction and Texture Space Rendering

To optimize Monte Carlo integration on this new hardware, Veach and Guibas [13] established the mathematical framework for combining sampling techniques, directly leading to the adoption of Next Event Estimation (NEE). NEE drastically reduces variance by explicitly casting a "shadow ray" directly toward the primary light source at every single bounce. This ensures rapid mathematical convergence, even in indirectly lit museum environments, by decoupling direct solar beams from diffuse ambient scattering.

Finally, to decouple this high-speed ray traversal from the constraints of the camera viewport, Munkberg et al. [14] introduced modern texture space caching methodologies. Standard screen-space renderers discard critical lighting data that falls outside the immediate view frustum. By shading directly into the unwrapped UV coordinates of an object, rendering becomes entirely independent of camera occlusion. This paradigm guarantees

persistent and accurate radiometric accumulation across the entire geometric topology of an artifact.

3.3.4. Solar Telemetry and Ephemeris Models

A rigorous dosimetric simulation, especially one simulating an entire meteorological year (8,760 hours), requires exact astronomical positioning of the sun. The solar position dictates the angle of direct irradiance, which directly influences both the length of shadows cast across the artifacts and the intensity of light penetrating through windows.

To calculate this, scientific engines utilize mathematical solar ephemeris models, such as the Solar Position Algorithm (SPA) developed by Reda and Andreas [15]. By inputting the geographic Latitude and Longitude of the heritage site, alongside the specific Julian Day of the year and the local time, the model computes the exact solar Zenith (altitude) and Azimuth angles.

Beyond accuracy, computing the solar telemetry is vital for computational optimization. Real-time path tracing is extraordinarily expensive. By calculating the solar altitude before executing the GPU ray-tracing kernel, the simulation can determine whether the sun is below the horizon. If the system detects nighttime hours (between sunset and sunrise), it can instantly skip the path-tracing execution for that hour. This astronomical culling mechanism drastically reduces the total computation time required to simulate an 8,760-hour cycle, bypassing thousands of redundant calculations without sacrificing metrological integrity.

3.3.5. The Perez All-Weather Sky Model

A virtual simulation of environmental dosimetry is only as accurate as the external light source driving it. While standard 3D rendering often utilizes a simple directional light to represent the sun and a uniform color for the sky, real-world atmospheric scattering is immensely complex. To address this, scientific metrology relies on rigorous atmospheric mathematics, specifically the Perez All-Weather Sky Model [12].

The Perez model is an empirical mathematical framework designed to accurately calculate the relative luminance of any point in the sky dome based on real-world meteorological data. Unlike idealized models that only simulate perfectly clear or completely overcast skies, the Perez model utilizes two specific atmospheric parameters:

- **Clearness (ϵ):** A metric defining the ratio of direct solar radiation to diffuse sky radiation, effectively quantifying the presence and thickness of cloud cover.
- **Brightness (Δ):** A metric describing the overall luminous opacity of the atmosphere, influenced by aerosol density and humidity.

By inputting these parameters alongside the precise solar altitude and azimuth, the model generates a highly accurate continuous luminance distribution. It distinctly simulates two critical atmospheric scattering phenomena: *circumsolar brightening* (the intense scattering of light in the immediate vicinity of the solar disk due to Mie scattering) and *horizon brightening* (the luminous gradient caused by Rayleigh scattering through the thicker atmosphere at the horizon). In the context of *ChronoLux*, when a path-traced ray escapes the building's geometry via the DXR Miss shader, its directional vector is passed directly into the Perez mathematical function, returning a physically accurate spectral radiance value for that exact micro-angle of the sky dome.

Conclusion: The methodologies reviewed in this chapter demonstrate that while offline renderers (Radiance) are accurate and IoT sensors are empirical, neither supports

interactive prediction. By synthesizing the mathematical rigor of Monte Carlo Path Tracing, NEE [13], and the Perez Sky Model [12] with the massive hardware acceleration of DXR [11], BVH traversal, and Texture Space Ray Tracing [14], the proposed *ChronoLux* framework bridges the critical gap in modern preventive conservation.

Chapter 4. Analysis and Theoretical Foundation

Together with the next two chapters takes about 70% of the whole paper.

The purpose of this chapter is to explain the operating principles of the implemented application. Here you write about your solution from a theory standpoint – i.e. you explain it and demonstrate its theoretical properties/value, e.g.:

- used or proposed algorithms,
- used protocols,
- abstract models,
- logic explanations/arguments concerning the chosen solution,
- logic and functional structure of the application, etc.

YOU SHOULD NOT write about the implementation. YOU SHOULD NOT describe technologies and other things which do not pertain to your project (no fillers, please!).

4.1. A section title

4.1.1. A Subsection title

Every table in the thesis should be numbered as Table $x.y$, where x is the chapter number where the table is included, and y is the number of the table within that chapter. There should be one empty line after the paragraph preceding the table, and one empty line after the table. Example: in this row we have inserted a reference to Table 4.1.

Table 4.1: Results

Case	Method#1	Method#2	Method#3
1	50	837	970
2	47	877	230
3	31	25	415

Every figure used in the thesis should be referred (e.g. Figure $x.y$ shows the components of the system...) and numbered.

Numbering is like this: Figure $x.y$ where x is the chapter number and y is the number of the figure within that chapter. E.g. , in this row we have inserted a reference to Figure 4.1.

Figure 4.1: Figure name

Chapter 5. Detailed Design and Implementation

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The purpose of this chapter is to document the developed application such a way that it can be maintained and developed later. A reader should be able (from what you have written here) to identify the main functions of the application.

The chapter should contain (but not limited to):

- a general application sketch/scheme,
- a description of every component implemented, at module level,
- class diagrams, important classes and methods from key classes.

Chapter 6. Testing and Validation

Together with the previous two chapters should take about 70% of the paper.

Chapter 7. User's Manual

In the section describing the installation procedure you should detail the hardware and software resources needed for installing and running the application, and a step by step description of how your application can be deployed/installed. An administrator should be able to perform the installation/deployment based on your instructions.

In the section for the user you should describe how to use the application from the point of view of a user with no inside technical information; this should be adorned with screen shots and a stepwise explanation of the interaction. Based on user's manual, a person should be able to install and use your product.

Should take 1 to 5 pages.

Chapter 8. Conclusions

This chapter should take one or two pages.

In this chapter you should include::

- A summary of your contributions/achievements,
- A critical analysis of the results achieved,
- A description of the possibilities of improvements/further development.

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Appendix A. Relevant Code sections

```
/** Maps are easy to use in Scala. */
object Maps {
  val colors = Map("red" -> 0xFF0000,
                   "turquoise" -> 0x00FFFF,
                   "black" -> 0x000000,
                   "orange" -> 0xFF8040,
                   "brown" -> 0x804000)

  def main(args: Array[String]) {
    for (name <- args) println(
      colors.get(name) match {
        case Some(code) =>
          name + " has code: " + code
        case None =>
          "Unknown color: " + name
      }
    )
  }
}
```

Appendix B. Other Relevant Info

Proofs etc. if any. Otherwise remove this chapter

Appendix C. Published Papers

If any. Otherwise remove this chapter